

VM2: Authentication Server

This virtual machine (VM2) is responsible for verifying user credentials (authentication). It acts as a **JSON-RPC** server on the network, responding to login requests sent by the web service `.(VM1`

Implemented Features

- Communication Protocol:** Utilizes `net/rpc/jsonrpc` for seamless communication with the `.web` service
- High Security (Design Bonus):** Passwords are not stored in plain text. All passwords in the database are hashed using the **SHA-256** algorithm. During a login attempt, the service `.hashes` the incoming password and compares it against the stored hash
- Data Isolation:** The database file (`users.json`) resides exclusively on this VM. The web service (VM1) has no direct access to read this file, adhering strictly to the separation of `.concerns`

File Structure

- `.main.go`: Contains the main RPC server initialization and authentication logic
- `users.json`: The text-based database storing usernames and their corresponding hashed `.passwords`

Test User For VM1 Login

(Password (Plain Text	Username
alice123	alice
bob123	bob
admin123	admin

Prerequisites

- `.(Go` installed on the operating system (Linux
- `.Ensure` port 8082 is open on VM2's firewall to allow incoming TCP connections from VM1

How to Run

- Open a terminal and navigate to the service directory: `.1`
`/cd auth-vm`

Run the service using Go: .2
go run main.go

Upon successful execution, the following message will appear in the terminal, indicating .3
the server is listening for requests:
VM2 (Auth RPC Server) is running on port 8082...
.Security Note: Using SHA-256 password hashing. Users loaded into memory

Networking Notes for Testing

To test the full flow, ensure you replace the target IP in your web service (VM1) code with the actual IP address of this machine (VM2). This ensures VM1 can successfully route its Login RPC .calls to port 8082 of VM2